

2015-2016 – Econ590
Topics in Macroeconomics

Lecture 0 : Business Cycle Facts and Detrending

Paul Beaudry & Franck Portier
franck.portier@TSE-fr.eu

Vancouver School of Economics

Version 1.0
20/01/2016

Disclaimer

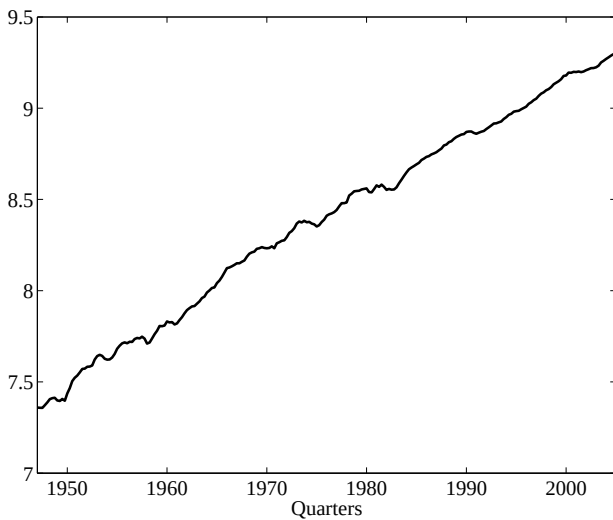
These are the slides I am using in class. They are not self-contained, do not always constitute original material and do contain some “cut and paste” pieces from various sources that I am not always explicitly referring to (not on purpose but because it takes time). Therefore, they are not intended to be used outside of the course or to be distributed. Thank you for signaling me typos or mistakes at franck.portier@TSE-fr.eu.

Introduction

- ▶ Macroeconomics is about the determination of aggregate variables, as measured by national accounts (output, consumption, employment, inflation,...)
- ▶ Economists makes a distinction (at least at first pass) between the long run and the short run, between Growth and Business Cycle
- ▶ For the methodological part of that lecture, I will consider the U.S.A. as an example.

Introduction

FIGURE 1 – US log Real GDP per capita



Introduction

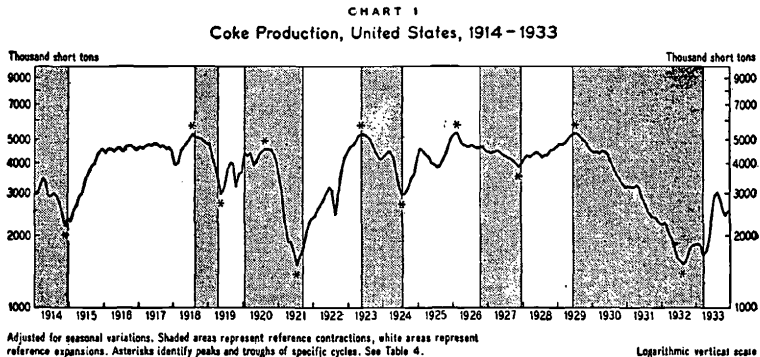
- ▶ Burns and Mitchell *“Measuring Business Cycles”* (1946, National Bureau of Economic Research) :

“Business cycles are a type of fluctuation found in the aggregate economic activity of nations that organize their work mainly in business enterprises : a cycle consists of expansions occurring at about the same time in many economic activities, followed by similarly general recessions, contractions, and revivals which merge into the expansion phase of the next cycle.”

- ▶ To identify cycles, B&M assume that they are no shorter than 6 quarters, and found a maximum length of 32 quarters.

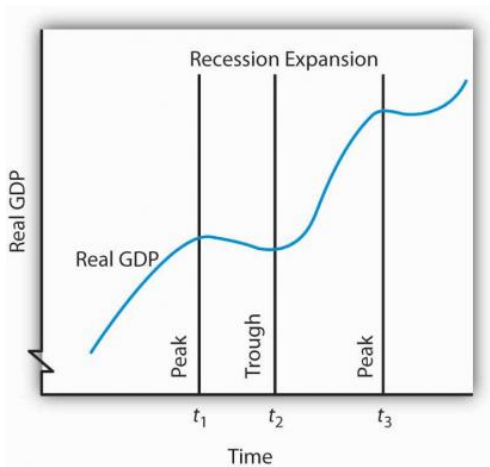
Introduction

FIGURE 2 – Reproduced from Burns and Mitchell *Measuring Business Cycles* (1946)



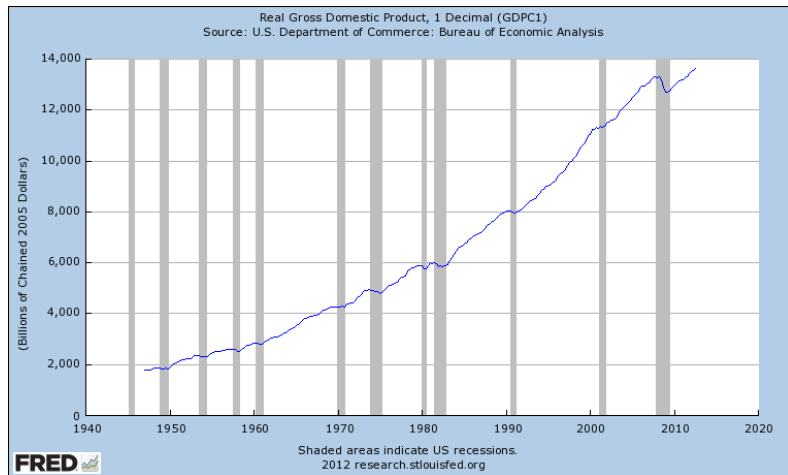
Introduction

FIGURE 3 – A Business Cycle



Introduction

FIGURE 4 – U.S. Business Cycles, as identified by the NBER's Business Cycle Dating Committee



Introduction

TABLE 1 – Recent U.S. Business Cycles, as identified by the NBER's Business Cycle Dating Committee

<u>BUSINESS CYCLE</u>		<u>DURATION IN MONTHS</u>			
<u>REFERENCE DATES</u>					
Peak	Trough	Contraction	Expansion	Cycle	
<i>Quarterly dates are in parentheses</i>		<i>Peak to Trough</i>	<i>Previous trough to this peak</i>	<i>Trough from Previous Trough</i>	<i>Peak from Previous Peak</i>
April 1960(II)	February 1961 (I)	10	24	34	32
December 1969(IV)	November 1970 (IV)	11	106	117	116
November 1973(IV)	March 1975 (I)	16	36	52	47
January 1980(I)	July 1980 (III)	6	58	64	74
July 1981(III)	November 1982 (IV)	16	12	28	18
July 1990(III)	March 1991(I)	8	92	100	108
March 2001(I)	November 2001 (IV)	8	120	128	128
December 2007 (IV)	June 2009 (II)	18	73	91	81

Introduction

TABLE 2 – Average Length of Expansions and Recessions for the U.S. Business Cycles (from the NBER) (in month)

	Contraction	Expansion	Cycle	
	P to T	T to P	T to T	P to P
1854-2009 (33 cycles)	16	42	56	55
1854-1919 (16 cycles)	22	27	48	49
1919-1945 (6 cycles)	18	35	53	53
1945-2009 (11 cycles)	11	59	73	66

Introduction

- ▶ How does the NBER establish this chronology ?
- ▶ Here is the “Statement of the NBER Business Cycle Dating Committee on the Determination of the Dates of Turning Points in the U.S. Economy” .

“The NBER’s Business Cycle Dating Committee maintains a chronology of the U.S. business cycle. The chronology comprises alternating dates of peaks and troughs in economic activity. A recession is a period between a peak and a trough, and an expansion is a period between a trough and a peak. During a recession, a significant decline in economic activity spreads across the economy and can last from a few months to more than a year. Similarly, during an expansion, economic activity rises substantially, spreads across the economy, and usually lasts for several years.”

“In both recessions and expansions, brief reversals in economic activity may occur – a recession may include a short period of expansion followed by further decline; an expansion may include a short period of contraction followed by further growth. The Committee applies its judgment based on the above definitions of recessions and expansions and has no fixed rule to determine whether a contraction is only a short interruption of an expansion, or an expansion is only a short interruption of a contraction . The most recent example of such a judgment that was less than obvious was in 1980-1982, when the Committee determined that the contraction that began in 1981 was not a continuation of the one that began in 1980, but rather a separate full recession.”

Introduction

- ▶ Is there a way to translate this into some statistical procedure?
- ▶ What are the data that we shall use and how are they constructed?
- ▶ What are the empirical regularities of BC?

1. A First Look at Some Methods To Extract the Cycle

- ▶ Any time series $y_t = \log Y_t$ can be decomposed such that

$$y_t = y_t^T + y_t^C$$

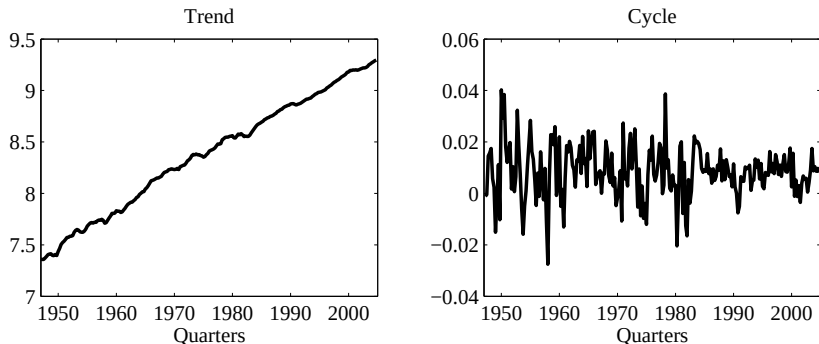
- ▶ Problem : How to define/identify each component ?
- ▶ Several ways of approaching the problem
- ▶ Actually : Infinite number of decomposition of a non-stationary process into a cycle and a trend
- ▶ Let us see some “intuitive” definition of those decompositions

1. A First Look at Some Methods To Extract the Cycle

Growth Cycle

- ▶ Take the growth rate of the series
- ▶ Expansion : Positive rate of growth
- ▶ Note : the cycle is very volatile (almost iid) – a lot of medium run fluctuations are eliminated

FIGURE 5 – US Growth Cycles



1. A First Look at Some Methods To Extract the Cycle

Trend Cycle

- ▶ Deviation from linear trend
- ▶ The trend is obtained from linear regression

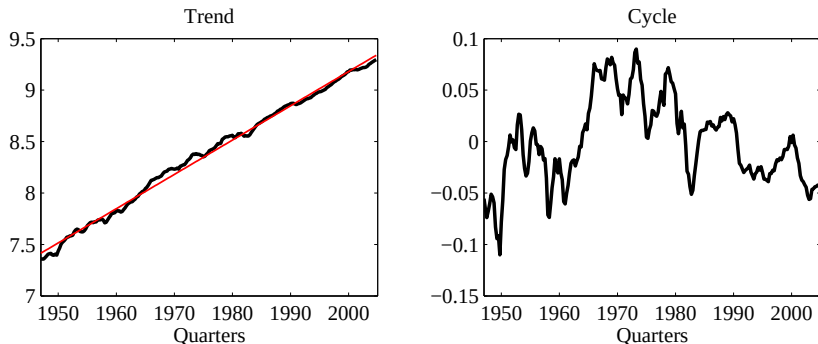
$$y_t = \alpha + \beta t + u_t$$

- ▶ Cycle : $y_t^C = y_t - (\hat{\alpha} + \hat{\beta}t)$
- ▶ Expansion : Output above the trend

1. A First Look at Some Methods To Extract the Cycle

Trend Cycle

FIGURE 6 – US Trend Cycles



- Note : the cycle can be large and very persistent - a lot of medium and long run fluctuations are not eliminated

1. A First Look at Some Methods To Extract the Cycle

Cycle = Output Gap

- ▶ Define the Output gap as

Actual output – Potential Output

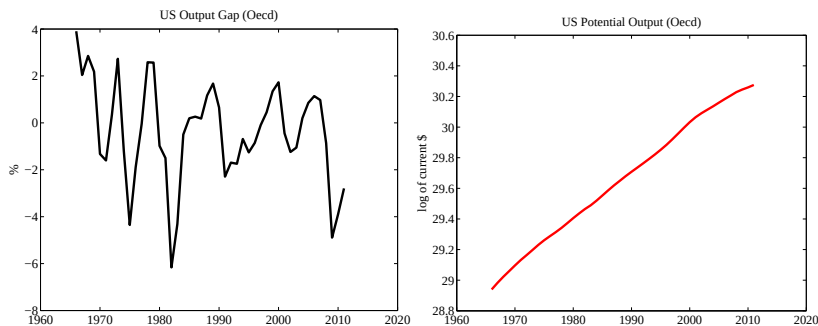
- ▶ Expansion : Actual output $\geq$ Potential output
- ▶ Actual output : easy to observe
- ▶ Note : How to identify potential output ? (full utilization ?, efficient ?)
- ▶ Example :
 1. estimate $y_t = \alpha \times u_t + \text{other controls} + \varepsilon_t$,
 2. define potential output as $y_t^P = \hat{\alpha} \times 0\% + \text{other controls} + \hat{\varepsilon}_t$.
(One might choose $u = u^n$ where u^n is the *natural* rate of unemployment (the Oecd chooses the NAIRU (Non Accelerating Inflation Rate of Unemployment)))
- ▶ Cycle is then $y_t - y_t^P$

1. A First Look at Some Methods To Extract the Cycle

Cycle = Output Gap

- This is an over simplified description of the method used by Oecd.

FIGURE 7 – US Output Gap and Potential Output



1. A First Look at Some Methods To Extract the Cycle

- ▶ One could describe many other methods to extract the Business Cycle.
- ▶ Ideally, we want to get rid of very short run and long run movements of economic activity.
- ▶ The best way to understand this is to decompose economic time series into the frequency domain and filter them.
- ▶ For this, we need to understand how a time series can be represented in the frequency domain.
- ▶ Here I am giving the main intuitions, a more rigorous treatment will be done in an econometrics course.

2. Decomposing a time series into frequency domain

Background

- ▶ A time series x_t is a collection of random variables
- ▶ It is co-variance stationary if its expectation and autocovariance does not depend on time
- ▶ A co-variance stationary process is therefore summarized by its auto-variance function
- ▶ Alternative representation : Moving average (link to impulse response)
- ▶ Wold Th : Every co-stationary times series have a (possibly infinite) moving average representation
- ▶ Spectral representation is an alternative representation, which is especially useful when discussing trend removal

2. Decomposing a time series into frequency domain

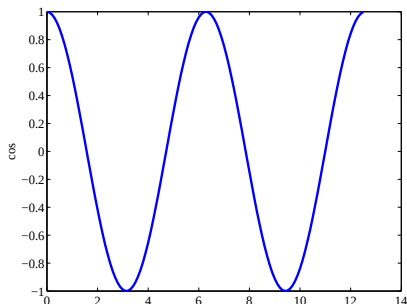
Typical periodic functions

- ▶ Idea : A series can be seen as the sum of periodic functions.
- ▶ A typical periodic function is $\cos(\omega t)$, with period (the time it takes to reproduce itself) $2\pi/\omega$.
 - × Knowing that period of $\cos(t)$ is 2π , for a given t_1 , what is the t_2 such that $\cos(\omega t_2) = \cos(\omega t_1)$?
 - × The solution is $t_2 - t_1 = 2\pi/\omega$.
- ▶ $\frac{\omega}{2\pi}$ is the *frequency* of oscillation (number of cycles per unit of time)

2. Decomposing a time series into frequency domain

Typical periodic functions

FIGURE 8 – Cosine wave with $\omega=1$

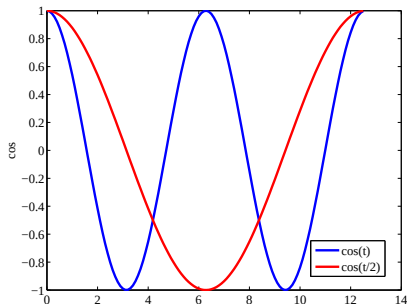


- With $\omega = 1$, the period is $2\pi = 6.28$ and frequency is $\frac{1}{2\pi} = 0.16$.

2. Decomposing a time series into frequency domain

Typical periodic functions

FIGURE 9 – Cosine waves with $\omega=1$ or $1/2$

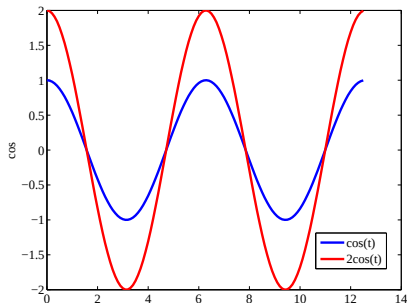


- With $\omega = 1/2$, the period is $4\pi = 12.56$ and frequency is $\frac{1}{4\pi} = 0.08$.

2. Decomposing a time series into frequency domain

Typical periodic functions

FIGURE 10 – Cosine waves with $\omega=1$ and different amplitudes



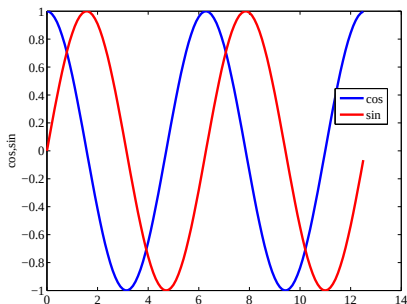
- Here are plotted $A \cos(t)$ with $A = 1$ or $A = 2$.

2. Decomposing a time series into frequency domain

Typical periodic functions

- $\sin(\omega t)$ behaves the same way, with same amplitude and period, but with a phase shift

FIGURE 11 – Cosine and Sine waves with $\omega=1$



2. Decomposing a time series into frequency domain

Typical periodic functions

- ▶ The idea of spectral decomposition is that with sin and cos, we can span the whole space of covariance stationary time series : the typical periodic function is

$$a \cos(\omega t) + b \sin(\omega t) \quad (1)$$

whose period is $2\pi/\omega$ but whose phase and amplitude depend on (a, b)

- ▶ Here we want to treat a and b as mean zero random variables.
- ▶ There is always a sum of type (1) periodic functions that reproduces a given time series
- ▶ The spectral density or *spectrum* of a series indicates the weight of each frequency (from low to high) in the total variance of the series.

2. Decomposing a time series into frequency domain

An approximation of the spectrum

- ▶ Assume that we observe y_t over T (even) periods, and that it is centered.
- ▶ Our goal is to decompose y_t into $T/2$ periodic functions of frequencies $\omega_1, \omega_2, \dots, \omega_{T/2}$, with

$$\omega_j = \frac{2\pi j}{T}, \quad j = 1, \dots, T/2$$

- ▶ Then, we want to write y_t as

$$\begin{aligned} y_t &= a_1 \cos(\omega_1 t) + b_1 \sin(\omega_1 t) \\ &+ a_2 \cos(\omega_2 t) + b_2 \sin(\omega_2 t) \\ &+ \dots \\ &+ a_{T/2} \cos(\omega_{T/2} t) + b_{T/2} \sin(\omega_{T/2} t) \end{aligned} \tag{2}$$

for $t=1, \dots, T$.

2. Decomposing a time series into frequency domain

An approximation of the spectrum

- Find the T parameters (a_i, b_i) in (2) by OLS.

$$X = \begin{bmatrix} \cos(\omega_1) & \sin(\omega_1) & \cdots & \cos(\omega_{T/2}) & \sin(\omega_{T/2}) \\ \cos(2\omega_1) & \sin(2\omega_1) & \cdots & \cos(2\omega_{T/2}) & \sin(2\omega_{T/2}) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \cos(T\omega_1) & \sin(T\omega_1) & \cdots & \cos(T\omega_{T/2}) & \sin(T\omega_{T/2}) \end{bmatrix}$$
$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ \vdots \\ y_{T-1} \\ y_T \end{bmatrix} \quad \beta = \begin{bmatrix} a_1 \\ b_1 \\ \vdots \\ \vdots \\ a_{T/2} \\ b_{T/2} \end{bmatrix}$$

- If we assume $Y = X\beta + u$, we can compute the a and b .

2. Decomposing a time series into frequency domain

An approximation of the spectrum

- ▶ The a and b coefficients are computed as

$$\beta = (X'X)^{-1}X'Y$$

- ▶ Given that we have T explanatory variables for T observations, the R^2 is one and $u = 0$. Here we are just solving a *representation* problem, not an estimation one.
- ▶ Note that the last column of X is a column of 0. It is replaced by a column of 1 to deal with non-centered series.

2. Decomposing a time series into frequency domain

An approximation of the spectrum

- The coefficient are given by

$$a_j = \frac{2}{T} \sum_{t=1}^T \cos(\omega_j t) y_t \quad (3)$$

$$b_j = \frac{2}{T} \sum_{t=1}^T \sin(\omega_j t) y_t \quad (4)$$

for $j \leq T/2 - 1$, and

$$a_{T/2} = \frac{1}{T} \sum_{t=1}^T \cos(\omega_{T/2} t) y_t \quad (5)$$

$$b_{T/2} = \frac{1}{T} \sum_{t=1}^T y_t \quad (6)$$

2. Decomposing a time series into frequency domain

An approximation of the spectrum

- ▶ This is of course an approximation of the series.
- ▶ The spectral representation is :

$$y_t = \int_0^\pi (a(\omega) \cos(\omega t) + b(\omega) \sin(\omega t)) d\omega \quad (7)$$

- ▶ Any covariance stationary times series process can be represented in the form of (7).
- ▶ The function that gives the variance of $a(\omega)$ and $b(\omega)$, as a function of ω is the spectrum.

2. Decomposing a time series into frequency domain

Extracting the business cycle (BC) component using the frequency domain representation

- ▶ A representation like (7) allow us to make precise the notion of extracting the business cycle component of y_t .
- ▶ Assume y_t is observed on a quarterly basis, and that the BC is defined as fluctuations of periods between 6 and 32 quarters (1.5 to 8 years), i.e. for $\omega \in [\underline{\omega} \ \overline{\omega}] = [\frac{2\pi}{32}, \frac{2\pi}{6}]$.
- ▶ The the BC component of y , denoted y^C , is

$$y_t^C = \int_{\underline{\omega}}^{\overline{\omega}} (a(\omega) \cos(\omega t) + b(\omega) \sin(\omega t)) d\omega \quad (8)$$

- ▶ It can be shown that this spectral representation has a time-series equivalent, which is an infinite two-sided moving average of y_t :

$$y_t^C = B_0 y_t + B_1 (y_{t-1} + y_{t+1}) + B_2 (y_{t-2} + y_{t+2}) + \cdots \quad (9)$$

$$\text{with } B_0 = \frac{\overline{\omega} - \underline{\omega}}{\pi} \text{ and } B_j = \frac{\sin(\overline{\omega}j) - \sin(\underline{\omega}j)}{\pi j}$$

2. Decomposing a time series into frequency domain

Extracting the business cycle (BC) component using the frequency domain representation

$$y_t^C = \int_{\underline{\omega}}^{\overline{\omega}} (a(\omega) \cos(\omega t) + b(\omega) \sin(\omega t)) d\omega \quad (8)$$

$$y_t^C = B_0 y_t + B_1(y_{t-1} + y_{t+1}) + B_2(y_{t-2} + y_{t+2}) + \dots \quad (9)$$

- ▶ Why things are not as simple as they look ?
- ▶ We have to make an approximation of (9) because it requires an infinity of observations.
- ▶ Therefore, the MA is truncated according to some distance criterium

2. Decomposing a time series into frequency domain

Filters

- We consider here *univariate stationary processes*.

Definition 1

The *autocovariance* of a series Y_t is defined as

$\lambda_\tau = \text{cov}(Y_t, Y_{t-\tau}) = E(Y_t Y_{t-\tau})$ with the assumption $E(Y_t) = 0$;.

Definition 2

For a sequence $a_0, a_1, \dots, a_j, \dots$, the **generating function** of this sequence is $a(z) = \sum_j a_j z^j$.

- Note : z needs not to have any interpretation
- The generating function (or z -transform) of a process Y_t is $Y(z) = \sum_t Y_t z^t$.

2. Decomposing a time series into frequency domain

Filters

- ▶ Correspondence between spectrum and auto-covariogram :
- ▶ Assume $z = e^{-i\omega} = \cos \omega - i \sin \omega$ and define

$$s(\omega) = \frac{1}{2\pi} \lambda(z) = \frac{1}{2\pi} \sum_{\tau=-\infty}^{+\infty} \lambda_{\tau} e^{-i\omega\tau}$$

- ▶ Then one can show that

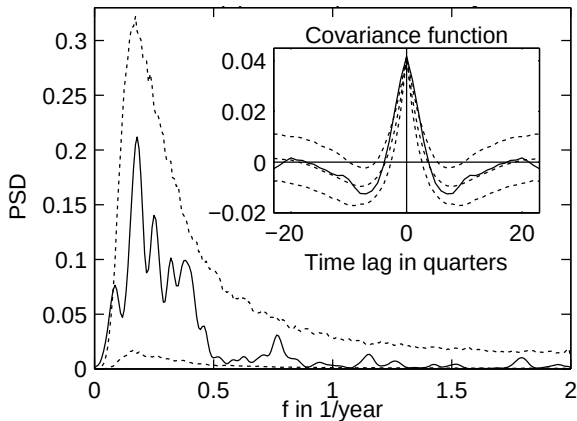
$$\lambda_{\tau} = \int_{-\pi}^{\pi} e^{i\tau\omega} s(\omega) d\omega \quad \text{and} \quad \lambda_0 = \int_{-\pi}^{\pi} s(\omega) d\omega$$

- ▶ The sequence of $\{\lambda_{\tau}\}$ and $\{s(\omega)\}$ bring the same information.
- ▶ The function $\frac{s(\omega)}{\lambda_0}$ has the property of a probability function over $-\pi \leq \omega \leq \pi$: $s(\omega) \geq 0$ and $\int_{-\pi}^{\pi} \frac{s(\omega)}{\lambda_0} d\omega = 1$.
- ▶ $s(\omega)$ (rescaled) is the spectral density.
- ▶ Next is an estimate of the spectral density, from GROTH, GHIL, HALLEGATTE and DUMAS, “Evidence from Genuine Periodicity and Deterministic Causes of US Business Cycles”, 2010.

2. Decomposing a time series into frequency domain

Filters

FIGURE 12 – A Estimate of US GDP Spectral Density (1954-2005, annual)



2. Decomposing a time series into frequency domain

Filters

Definition 3

Let $\hat{Y}_t = \sum_{j=0}^m c_j Y_{t-j} = C(L)Y_t$. $\hat{Y}_t$ is a **filtered** version of Y_t .

- One can show that the spectral density of $\hat{Y}_t$ is

$$s_{\hat{Y}}(\omega) = C(e^{-i\omega})C(e^{i\omega})s_Y(\omega) = |C(e^{i\omega})|^2 s_Y(\omega)$$

.

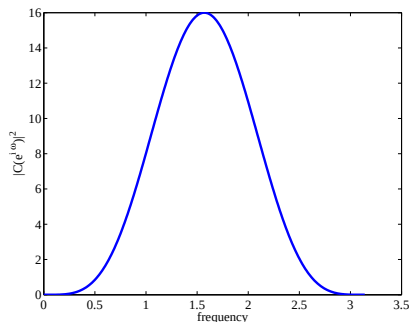
2. Decomposing a time series into frequency domain

Band Pass Filter

- ▶ $c_0 = 1, c_2 = -2, c_4 = -1$ and $c_j = 0$ for other j .
- ▶ $\hat{Y}_t = C(L)Y_t = Y_t - 2(Y_{t-2} + Y_{t+2}) - (Y_{t-4} + Y_{t+4})$
- ▶ Then

$$|C(e^{i\omega})|^2 = 4(1 - \cos 2\omega)^2$$

FIGURE 13 – A Band Pass Filter



2. Decomposing a time series into frequency domain

Band Pass Filter

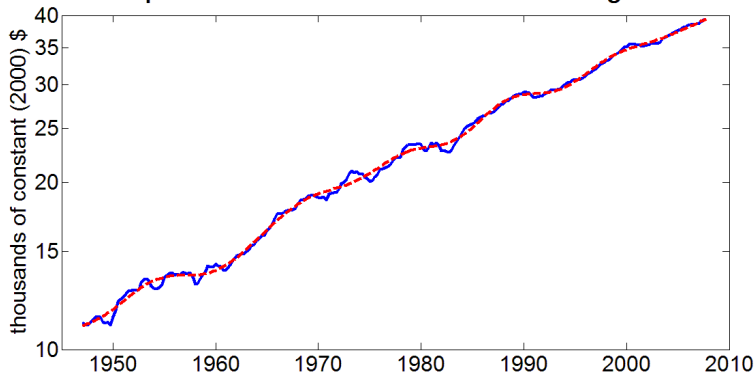
- ▶ Here is an example of the use of Band Pass filters from ROBERTO PANCRAZI, “Spectral Covariance Instability Test : An Application to the Great Moderation”, TSE 2010.
- ▶ High Frequency : periodicity between 2 and 32 quarters
- ▶ Medium Frequency : periodicity between 32 and 80 quarters
- ▶ High to Medium Frequency : periodicity between 2 and 80 quarters

2. Decomposing a time series into frequency domain

Band Pass Filter

FIGURE 14 – US GDP

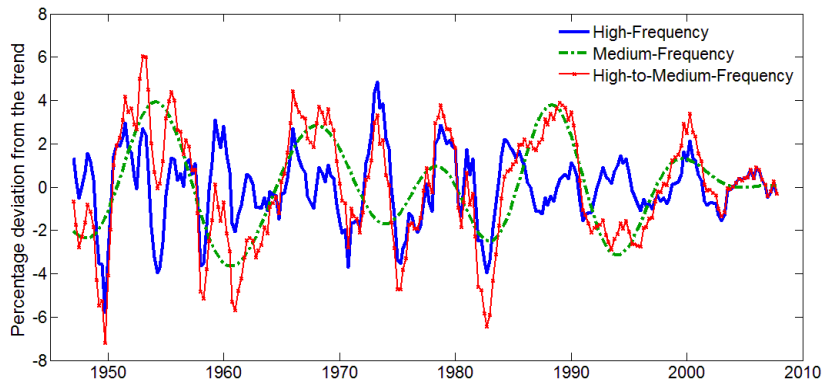
Real Per-Capita Gross Domestic Product on a Logarithmic Scale



2. Decomposing a time series into frequency domain

Band Pass Filter

FIGURE 15 – Various Cycles



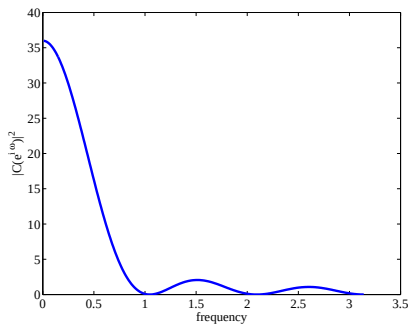
2. Decomposing a time series into frequency domain

Low Pass Filter

► $\tilde{Y}_t = \sum_{j=0}^m Y_{t-j}$. Then

$$|C(e^{i\omega})|^2 = \frac{1 - \cos(m+1)\omega}{1 - \cos \omega}$$

FIGURE 16 – A Low Pass Filter



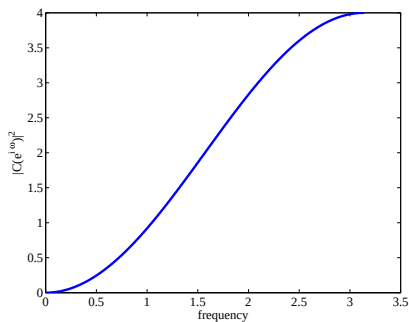
2. Decomposing a time series into frequency domain

First Difference

- First difference $\tilde{Y}_t = (1 - L)Y_t$. Then

$$|C(e^{i\omega})|^2 = 2 - 2\cos\omega$$

FIGURE 17 – First Difference



2. Decomposing a time series into frequency domain

A High Pass Filter : The HODDRICK-PRESCOTT Filter

- ▶ Very popular in the macro literature
- ▶ In the time domain, the idea is to remove a trend which is smooth, but not linear
- ▶ The trend y_t^T is the Argmin of :

$$\sum_{t=1}^T (y_t - y_t^T)^2 + \lambda \sum_{t=2}^T ((y_{t+1}^T - y_t^T) - (y_t^T - y_{t-1}^T))^2$$

- ▶ if $\lambda = +\infty$, it is linear detrending.
- ▶ The solution of this program solves
$$y_t/\lambda = y_{t+2}^T - 4y_{t+1}^T + (6 + \lambda)y_t^T - 4y_{t-1}^T - y_{t-2}^T$$
- ▶ The solution is a symmetric MA of order $+\infty$:

$$y_t^T = \sum_{j=-\infty}^{\infty} a_{|j|} y_{t+j}$$

- ▶ Then $y_t^C = y_t - y_t^T$ is a time invariant linear symmetric filter.

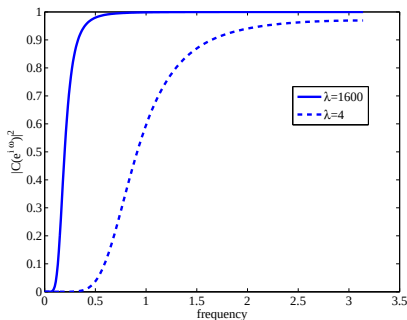
2. Decomposing a time series into frequency domain

A High Pass Filter : The HODDRICK-PRESCOTT Filter

- ▶ With $\lambda = 1600$ on quarterly data, it removes cycles of period greater than 10 years.
- ▶ The transfer function is

$$|C(e^{i\omega})|^2 = \frac{16\lambda^2(1 - \cos \omega)^4}{(1 + 4\lambda(1 - \cos \omega)^2)^2}$$

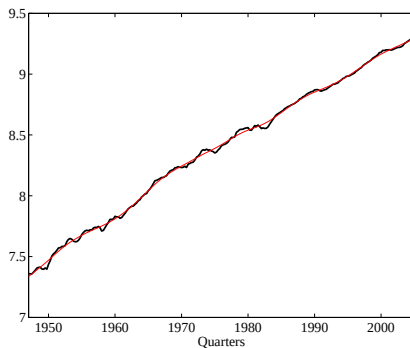
FIGURE 18 – HODDRICK-PRESCOTT Filter



2. Decomposing a time series into frequency domain

The HP filter at work

FIGURE 19 – US HP Trend



2. Decomposing a time series into frequency domain

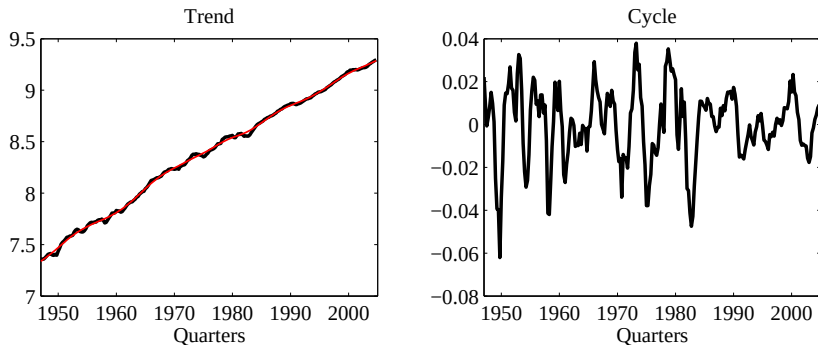
assignment 0

1. Get output data for the US and 1 other country.
2. Re-do the Pancrazi exercise for these two countries.
3. Compare the HP filter with a Band pass filter 2-32 for the two series
4. Do the Pancrazi exercise for US data on Investment, Total Consumption, Consumption services, Durable goods purchases, Equipment investment and structures investment.

2. Decomposing a time series into frequency domain

The HP filter at work

FIGURE 20 – US HP Cycle



4. U.S. Business Cycles

Business Cycles = Comovements

- ▶ Lucas' definition :

*“Recurrent **fluctuations** of macroeconomic aggregates around trend”*

- ▶ Want to find **regularities** (*Stylized facts*)

- ▶ Business Cycles are characterized by a set of statistics :

- × Volatilities of time series (standard deviations)
- × Comovements of time series (correlations, serial correlations)

- ▶ Why only looking at the US ?

“Though there is absolutely no theoretical reason to anticipate it, one is led by the facts to conclude that, with respect to the qualitative behavior of co-movements among series, business cycles are all alike.” (LUCAS 1977)

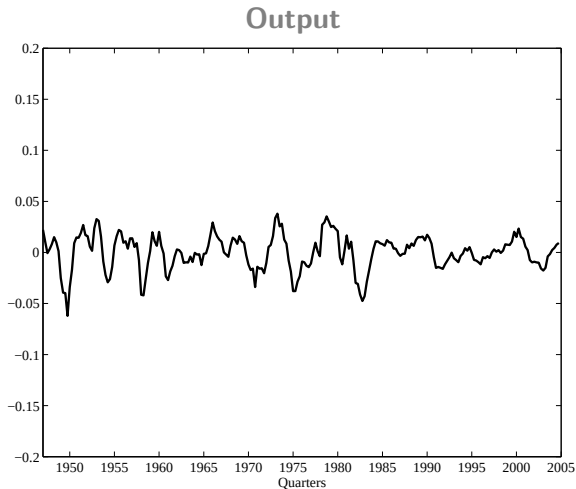
4. U.S. Business Cycles

Main Real Aggregates

- ▶ Consumption (C) : Nondurables + Services
- ▶ Investment (I) : Durables + Fixed Investment + Changes in inventories
- ▶ Government spending (G)
- ▶ Output : $C + I + G$
- ▶ Labor : hours worked
- ▶ Labor Productivity : Output / Labor

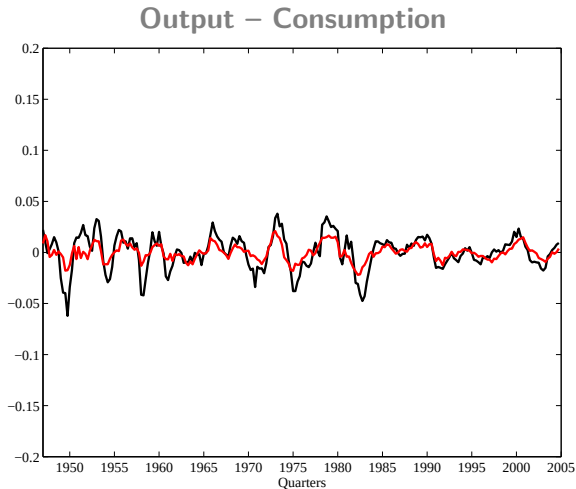
4. U.S. Business Cycles

Main Real Aggregates



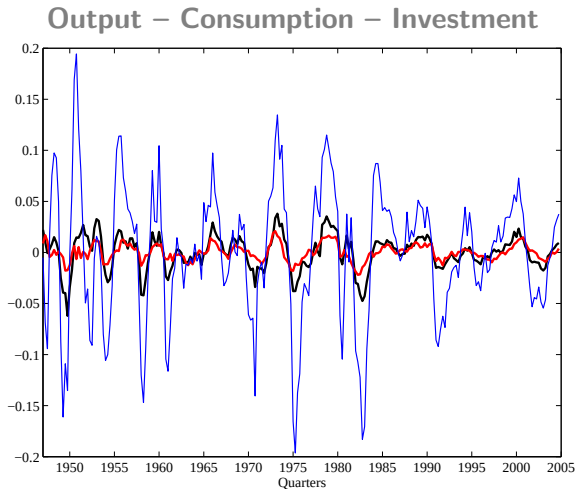
4. U.S. Business Cycles

Main Real Aggregates



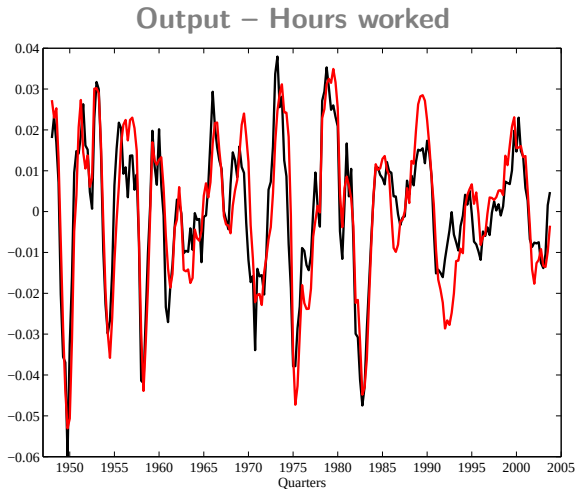
4. U.S. Business Cycles

Main Real Aggregates



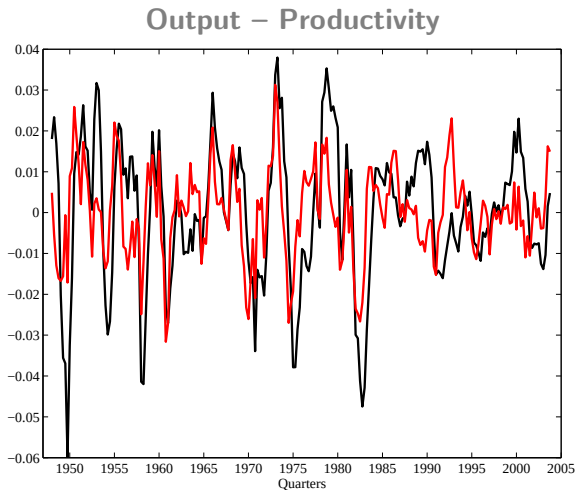
4. U.S. Business Cycles

Main Real Aggregates



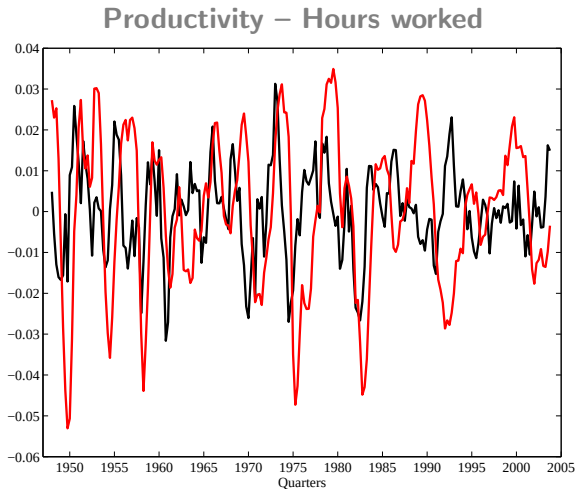
4. U.S. Business Cycles

Main Real Aggregates



4. U.S. Business Cycles

Main Real Aggregates



4. U.S. Business Cycles

More (Much More) Data

- ▶ Those figures are taken from STOCK and WATSON, “Business Cycle Fluctuations in US Macroeconomics Time Series”, chapter 1 of the Handbook of Macroeconomics, 1999
- ▶ Quarterly US data, 1947-1997

4. U.S. Business Cycles

More (Much More) Data

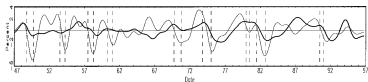


Fig. 3.3. Finance, insurance and real estate employment.

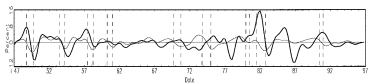


Fig. 3.4. Mining employment.

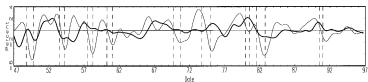


Fig. 3.5. Government employment.

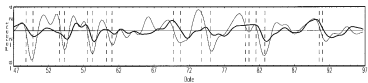


Fig. 3.6. Service employment.

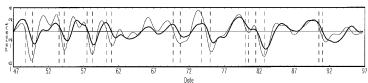


Fig. 3.7. Wholesale and retail trade employment.

4. U.S. Business Cycles

More (Much More) Data

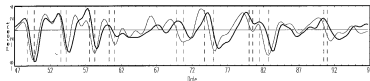


Fig. 3.8. Transportation and public utility employment.

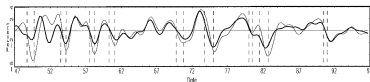


Fig. 3.9. Consumption (total).

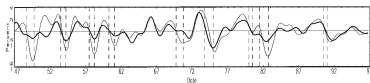


Fig. 3.10. Consumption (nondurables).

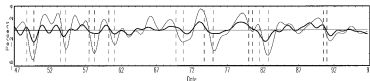


Fig. 3.11. Consumption (services).

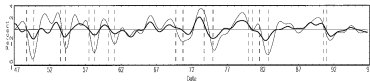


Fig. 3.12. Consumption (nondurables + services).

4. U.S. Business Cycles

More (Much More) Data

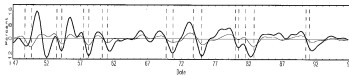


Fig. 3.13. Consumption (durables).

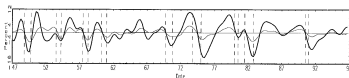


Fig. 3.14. Investment (total fixed).

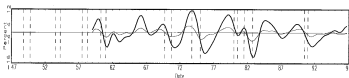


Fig. 3.15. Investment (equipment).

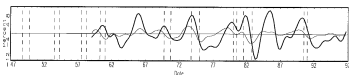


Fig. 3.16. Investment (nonresidential structures).

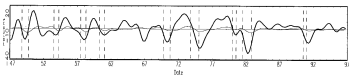


Fig. 3.17. Investment (residential structures).

4. U.S. Business Cycles

More (Much More) Data

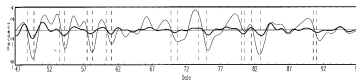


Fig. 3.18. Change in business inventories (relative to trend GDP).

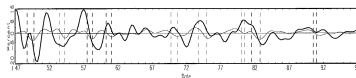


Fig. 3.19. Exports.

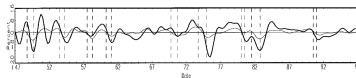


Fig. 3.20. Imports.

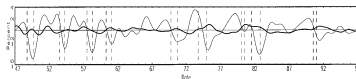


Fig. 3.21. Trade balance (relative to trend GDP).

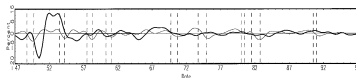


Fig. 3.22. Government purchases.

4. U.S. Business Cycles

More (Much More) Data

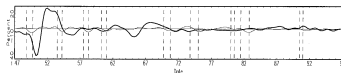


Fig. 3.23. Government purchases (defense).

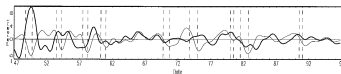


Fig. 3.24. Government purchases (non-defense).

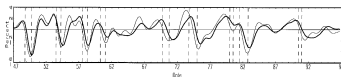


Fig. 3.25. Employment (total employees).

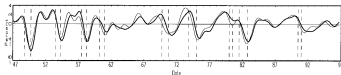


Fig. 3.26. Employment (total hours).

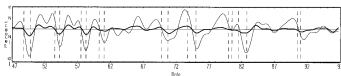


Fig. 3.27. Employment (average weekly hours).

4. U.S. Business Cycles

More (Much More) Data

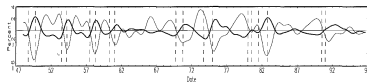


Fig. 3.28. Unemployment rate.

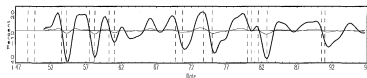


Fig. 3.29. Vacancies (Help Wanted index).

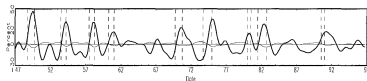


Fig. 3.30. New unemployment claims.

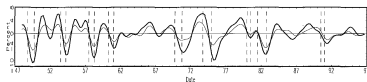


Fig. 3.31. Capacity utilization.

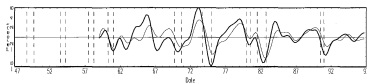


Fig. 3.32. Total factor productivity.

4. U.S. Business Cycles

More (Much More) Data

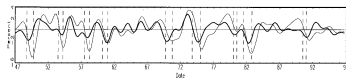


Fig. 3.33. Average labor productivity.

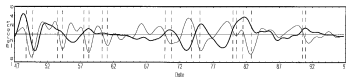


Fig. 3.34. Consumer price index (level).

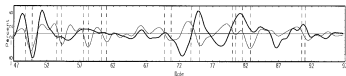


Fig. 3.35. Producer price index (level).

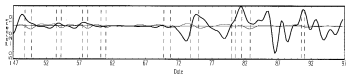


Fig. 3.36. Oil prices.

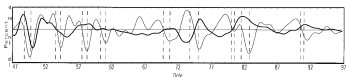


Fig. 3.37. GDP price deflator (level).

4. U.S. Business Cycles

More (Much More) Data

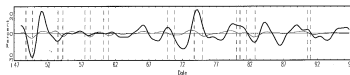


Fig. 3.38. Commodity price index (level).

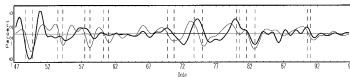


Fig. 3.39. Consumer price index (inflation rate).

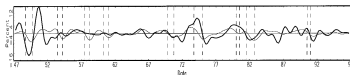


Fig. 3.40. Producer price index (inflation rate).

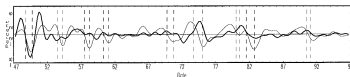


Fig. 3.41. GDP price deflator (inflation rate).

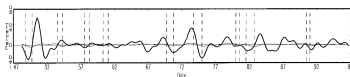


Fig. 3.42. Commodity price index (inflation rate).

4. U.S. Business Cycles

More (Much More) Data

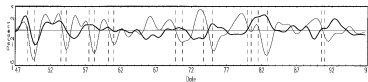


Fig. 3.43. Nominal wage rate (level).

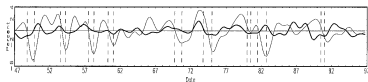


Fig. 3.44. Real wage rate (level).

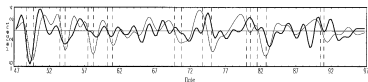


Fig. 3.45. Nominal wage rate (rate of change).

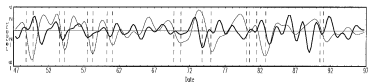


Fig. 3.46. Real wage rate (rate of change).

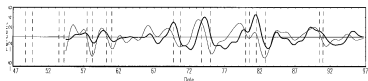


Fig. 3.47. Federal funds rate.

4. U.S. Business Cycles

More (Much More) Data

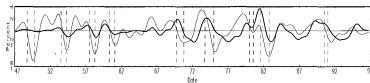


Fig. 3.48. Treasury Bill rate (3 month).

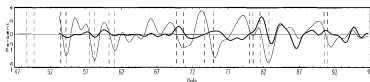


Fig. 3.49. Treasury Bond rate (10 year).

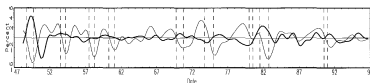


Fig. 3.50. Real Treasury Bill rate (3 month).

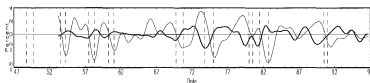


Fig. 3.51. Yield curve spread (long-short).

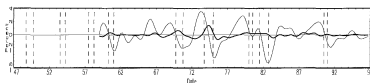


Fig. 3.52. Commercial paper/Treasury Bill spread.

4. U.S. Business Cycles

More (Much More) Data

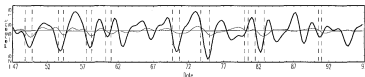


Fig. 3.53. Stock prices.

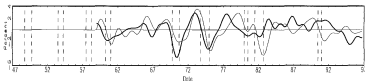


Fig. 3.54. Money stock (M2, nominal level).

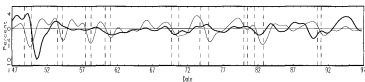


Fig. 3.55. Monetary base (nominal level).

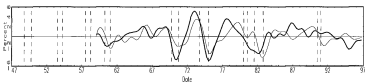


Fig. 3.56. Money stock (M2, real level).

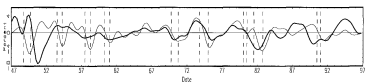


Fig. 3.57. Monetary base (real level).

4. U.S. Business Cycles

More (Much More) Data

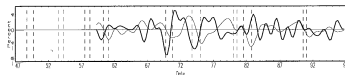


Fig. 3.58. Money stock (M2, nominal rate of change).

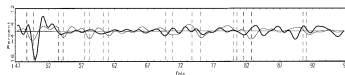


Fig. 3.59. Monetary base (Nominal rate of change).

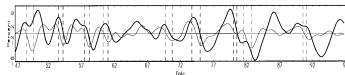


Fig. 3.60. Consumer credit.

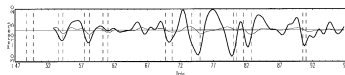


Fig. 3.61. Consumer expectations.

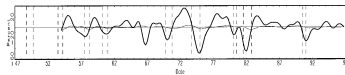


Fig. 3.62. Building permits.

4. U.S. Business Cycles

More (Much More) Data

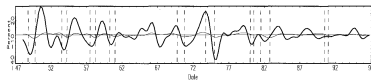


Fig. 3.63. Vendor performance.

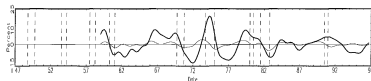


Fig. 3.64. Manufacturers' unfilled orders, durable goods industry.

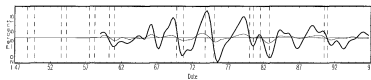


Fig. 3.65. Manufacturers' new orders, non-defense capital goods.

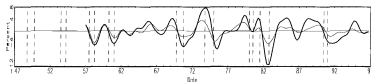


Fig. 3.66. Industrial production, Canada.

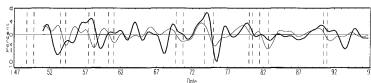


Fig. 3.67. Industrial production, France.

4. U.S. Business Cycles

More (Much More) Data

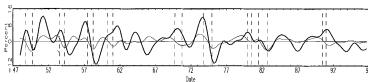


Fig. 3.68. Industrial production, Japan.

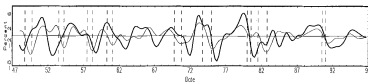


Fig. 3.69. Industrial production, UK.

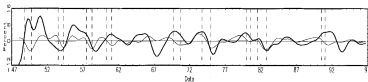


Fig. 3.70. Industrial production, Germany.

4. U.S. Business Cycles

Moments

- ▶ We want to characterize fluctuations $\rightsquigarrow$ amplitude and movements
- ▶ Amplitude : volatilities $\rightsquigarrow$ standard deviations
- ▶ Comovements : correlations

Variable	$\sigma(\cdot)$	$\sigma(\cdot)/\sigma(y)$	$\rho(\cdot, y)$	$\rho(\cdot, h)$	Serial(1)
Output	1.70	—	—	—	0.84
Consumption	0.80	0.47	0.78	—	0.83
<i>Services</i>	1.11	0.66	0.72	—	0.80
<i>Non Durables</i>	0.72	0.42	0.71	—	0.77
Investment	6.49	3.83	0.84	—	0.81
<i>Fixed investment</i>	5.08	3.00	0.80	—	0.88
<i>Durables</i>	5.23	3.09	0.58	—	0.72
<i>Changes in inventories</i>	22.48	13.26	0.48	—	0.40
Hours worked	1.69	1.00	0.86	—	0.89
Labor productivity	0.90	0.53	0.41	0.09	0.69

4. U.S. Business Cycles

Summary

1. Consumption (of non-durables) is less volatile than output
2. Investment is more volatile than output
3. Hours worked are as volatile as output
4. Capital is much less volatile than output
5. Labor productivity is less volatile than output
6. Real wage is much less volatile than output
7. All those variables are persistent and procyclical except Labor productivity that is acyclical

4. U.S. Business Cycles

Lucas

- ▶ Quoting LUCAS 1977 “Understanding Business Cycles”
 1. Output movements across broadly defined sectors move together.
 2. Production of producer and consumer durables exhibits much greater amplitude than does the production of nondurables
 3. Production and prices of agricultural goods and natural resources have lower than average conformity.
 4. Business profits show high conformity and much greater amplitude than other series.
 5. Prices generally are procyclical.
 6. Short-term interest rates are procyclical ; long-term rates slightly so.
 7. Monetary aggregates and velocity measures are procyclical.

5. International Business Cycles

Within countries

- ▶ From FIORITO and KOLLINTZAS, “Stylized facts of business cycles in the G7 from a real business cycles perspective”, European Economic Review, 1994.
- ▶ Quarterly data 1960-1989

5. International Business Cycles

Within countries

Cross correlations of real GNP/GDP with the components of spending, income, and output in levels. ^{a,b}

Variable	Vol.	X_{t-5}	X_{t-4}	X_{t-3}	X_{t-2}	X_{t-1}	X_t	X_{t+1}	X_{t+2}	X_{t+3}	X_{t+4}	X_{t+5}
(1) Real GNP/GDP												
USA	1.74	0.01	0.21	0.41	0.65	0.85	1.0	0.85	0.65	0.41	0.21	0.01
Canada	1.39	-0.12	0.04	0.27	0.51	0.78	1.0	0.78	0.51	0.27	0.04	-0.12
Japan	1.53	0.02	0.19	0.38	0.59	0.78	1.0	0.78	0.59	0.38	0.19	0.02
Germany	1.69	-0.02	0.23	0.35	0.46	0.67	1.0	0.67	0.46	0.35	0.23	-0.02
France	0.90	-0.06	0.10	0.30	0.54	0.77	1.0	0.77	0.54	0.30	0.10	-0.06
UK	1.54	-0.02	0.07	0.20	0.37	0.55	1.0	0.55	0.37	0.20	0.07	-0.02
Italy	1.70	-0.21	-0.04	0.22	0.52	0.80	1.0	0.80	0.52	0.22	-0.04	-0.21
(2) Consumption expenditure												
US	1.29	0.32	0.48	0.59	0.72	0.79	0.80	0.63	0.43	0.22	0.00	-0.17
Canada	1.27	-0.08	0.16	0.40	0.57	0.72	0.79	0.65	0.44	0.27	0.06	-0.03
Japan	1.33	-0.10	0.08	0.28	0.42	0.56	0.72	0.54	0.40	0.22	0.01	-0.11
Germany	1.53	0.11	0.26	0.37	0.46	0.58	0.69	0.55	0.49	0.38	0.32	0.21
France	0.86	-0.27	0.42	-0.63	0.73	0.72	0.62	0.30	0.10	-0.14	0.25	-0.32
UK	1.67	0.03	0.13	0.30	0.39	0.46	0.67	0.42	0.38	0.26	0.10	0.08
Italy	1.19	-0.15	0.07	0.34	0.59	0.74	0.78	0.69	0.50	0.25	0.03	-0.15
(3) Fixed investment												
US	5.57	0.14	0.30	0.47	0.67	0.83	0.90	0.78	0.59	0.35	0.12	-0.09
Canada	4.60	-0.43	-0.29	-0.07	0.18	0.40	0.53	0.52	0.41	0.32	0.21	0.14
Japan	4.57	-0.11	0.04	0.23	0.45	0.64	0.83	0.78	0.69	0.51	0.29	0.05

5. International Business Cycles

Within countries

Cross correlations of real GNP/GDP with the components of spending, income, and output in levels. ^{a,b}

Variable	Vol.	X_{t-5}	X_{t-4}	X_{t-3}	X_{t-2}	X_{t-1}	X_t	X_{t+1}	X_{t+2}	X_{t+3}	X_{t+4}	X_{t+5}
Germany	4.90	0.04	0.26	0.37	0.42	0.60	0.84	0.54	0.42	0.37	0.29	0.12
France	2.70	-0.11	0.06	0.26	0.46	0.66	0.78	0.69	0.57	0.41	0.25	0.13
UK	3.57	-0.11	-0.04	0.08	0.23	0.33	0.60	0.53	0.38	0.31	0.23	0.05
Italy	4.88	-0.16	-0.00	0.23	0.47	0.70	0.88	0.81	0.67	0.47	0.25	0.05
(5) Equipment investment												
US	6.28	-0.13	0.02	0.21	0.46	0.68	0.86	0.87	0.77	0.59	0.38	0.18
Canada	7.13	-0.49	-0.35	-0.18	0.03	0.27	0.43	0.51	0.53	0.50	0.34	0.25
Japan	5.96	-0.09	0.02	0.17	0.38	0.58	0.74	0.73	0.69	0.54	0.34	0.14
Germany	6.09	0.12	0.36	0.48	0.52	0.61	0.73	0.58	0.49	0.39	0.23	0.09
France	3.90	0.08	-0.23	0.39	0.58	0.70	0.74	0.53	0.31	0.12	-0.06	-0.17
UK	4.88	-0.12	-0.07	0.05	0.21	0.38	0.56	0.51	0.47	0.44	0.32	0.25
Italy	7.92	-0.15	0.01	0.25	0.48	0.69	0.85	0.74	0.57	0.38	0.14	-0.05
(6) Construction investment												
US	6.26	0.31	0.45	0.57	0.70	0.80	0.78	0.58	0.35	0.11	-0.10	-0.27
Canada	3.83	-0.23	-0.12	0.10	0.34	0.50	0.55	0.41	0.18	0.06	0.01	-0.04
Japan	5.58	-0.04	0.09	0.23	0.31	0.32	0.43	0.35	0.18	0.07	-0.05	-0.18
Germany	5.56	0.00	0.15	0.22	0.27	0.47	0.72	0.40	0.28	0.27	0.25	0.10
France	2.49	-0.25	-0.11	0.08	0.25	0.48	0.65	0.65	0.65	0.54	0.45	0.33
UK	3.90	0.15	0.19	0.28	0.26	0.21	0.38	0.27	0.08	-0.00	-0.08	-0.24
Italy	3.57	-0.11	0.00	0.18	0.36	0.57	0.74	0.74	0.65	0.50	0.36	0.20
(7) Inventory changes												
US	18.2	-0.01	0.08	0.22	0.35	0.49	0.64	0.48	0.26	0.03	-0.14	-0.30
Canada	35.4	0.07	0.15	0.25	0.43	0.60	0.68	0.53	0.33	0.06	-0.18	-0.32
Japan	45.4	-0.05	-0.03	0.07	0.23	0.38	0.38	0.38	0.25	0.20	0.20	0.10
Germany	49.2	0.07	0.19	0.31	0.32	0.33	0.35	0.29	0.14	0.02	-0.13	-0.27
France	30.1	-0.15	-0.09	-0.04	0.05	0.22	0.47	0.44	0.25	0.16	-0.05	-0.27
UK	26.6	0.03	0.12	0.16	0.26	0.42	0.55	0.38	0.19	0.00	-0.08	-0.17
Italy	66.8	-0.07	0.10	0.21	0.39	0.51	0.56	0.32	0.00	-0.24	-0.41	-0.48
(8) Government final consumption												
US	1.98	-0.07	-0.04	0.00	0.06	0.11	0.19	0.24	0.27	0.30	0.35	0.37
Canada	1.46	-0.18	-0.20	-0.24	-0.23	-0.20	-0.12	-0.09	-0.08	0.05	0.14	0.18
Japan	2.89	0.25	0.33	0.30	0.28	0.30	0.32	0.04	-0.05	-0.08	-0.05	-0.06
Germany	1.47	-0.19	-0.11	-0.13	-0.10	-0.06	0.05	0.06	0.16	0.23	0.36	0.41
France	0.70	0.46	0.61	0.56	0.46	0.32	0.18	-0.07	-0.23	-0.31	-0.30	-0.24
UK	1.43	-0.09	-0.03	-0.07	-0.06	0.02	0.04	-0.05	-0.01	-0.07	-0.05	0.04
Italy	0.60	0.30	0.18	0.05	-0.14	-0.30	-0.39	-0.43	-0.41	-0.33	-0.21	-0.04

5. International Business Cycles

Between countries

- ▶ The cross-correlations of output are high
- ▶ The cross-correlations of output are higher than the one of productivity
- ▶ The cross-correlations of productivity are higher than the cross-correlations of consumption.
- ▶ The cross-correlations of output, investment and employment are generally positive.
- ▶ See AMBLER, CARDIA and ZIMMERMANN, “International business cycles : What are the facts ?”, Journal of Monetary Economics, 2004.

5. International Business Cycles

Between countries

Average cross-correlations

Variable	Averages from 190 cross-correlations		
	Full sample		
	1960:1–2000:4	1973:1–2000:4	1973:1–1990:4
Output	0.22 (0.03) 0.00	0.28 (0.03) 0.00	0.30 (0.03) 0.00
Consumption	0.14 (0.02) 0.00	0.15 (0.03) 0.00	0.14 (0.03) 0.00
Investment	0.18 (0.04) 0.00	0.22 (0.04) 0.00	0.22 (0.03) 0.00
Employment	0.20 (0.03) 0.00	0.22 (0.03) 0.00	0.21 (0.04) 0.00
Total hours	0.26 (0.04) 0.00	0.29 (0.04) 0.00	0.26 (0.03) 0.00
Employment ^a	0.25 (0.04) 0.00	0.26 (0.04) 0.00	0.25 (0.05) 0.00
Productivity (from y and n only)	0.16 (0.02) 0.00	0.21 (0.02) 0.00	0.24 (0.03) 0.00
Productivity (best available) ^b	0.09 (0.02) 0.00	0.11 (0.02) 0.00	0.13 (0.02) 0.00